AD-Rank & Shape-CFD v4

AD-Rank & Shape-CFD: Physics-Inspired Reranking via Conservative Advection-Diffusion on Document Point Clouds for Dense Retrieval

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Abstract

Neural dense retrieval systems typically rank candidate documents by computing independent, pairwise cosine similarities between query and document embeddings, discarding the rich inter-document structural information that resides in the retrieved set. We present **Shape-CFD**, a physics-inspired reranking framework that recasts the post-retrieval reranking problem as a *conservative convection-diffusion process* on a sparse nearest-neighbor graph constructed over the candidate documents. Our formulation introduces two principal innovations: (1) an **advective transport term** that encodes query-directed momentum along graph edges, complementing the isotropic diffusion employed by prior graph-based rerankers; and (2) a **shape-aware extension** that upgrades each document from a single point embedding to a sentence-level *point cloud*, measuring inter-document affinity via the symmetric Chamfer distance. To mitigate the variance collapse of inner products in high-dimensional spaces ($d = 4096$), we apply Johnson-Lindenstrauss random projection to a compact $m = 128$ subspace, which simultaneously achieves a $32\times$ computational speedup and introduces a stochastic regularization effect analogous to simulated annealing. The resulting iterative scheme provably converges to a unique stationary distribution under a Courant-Friedrichs-Lewy (CFL) stability condition, and its conservative upwind discretization guarantees global mass conservation. On three BEIR benchmarks, Shape-CFD achieves competitive NDCG@10 scores—including a **+21.5%** improvement on NFCorpus—while maintaining a retrieval latency of **22 ms** on commodity CPU hardware, approximately $23\times$ faster than cross-encoder rerankers. **In this v3 supplement**, we report a systematic ablation of the Allen-Cahn reaction-diffusion extension (proposed as future work in v1) across NFCorpus and SciFact, demonstrate that nonlinear reaction dynamics are numerically dormant in the narrow score-range regime of short queries ($\Delta C_0 \approx 0.1$), and discover that a *mixed initial field* $C_0 = \alpha \cdot \text{MaxSim} + (1-\alpha) \cdot \text{MeanSim}$ with $\alpha = 0.7$ yields a **+3.2%** improvement over Shape-CFD v4 on NFCorpus—but degrades on SciFact. We formalize this dataset-dependent behavior as the **Pareto boundary of initial fields**, governed by the trade-off between extreme-value sampling noise and signal dilution, and propose a scale-adaptive symmetry-breaking formula that reduces to the original Chamfer baseline as a special case. Additionally, we introduce **PQ-Chamfer** (Subspace Chamfer Distance), a structured decomposition that partitions the 4096-dimensional embedding space into 64 contiguous 64-dimensional subspaces and computes independent cosine distances within each, yielding a **+8.8%** improvement over the full-dimensional baseline—compared to the −7.6% degradation of random JL projection—demonstrating that preserving dimensional structure

is critical for high-dimensional distance computation. **In this v4 supplement**, we discover that treating each PQ subspace as an independent “virtual token” and reversing the order of subspace aggregation and sentence-level Chamfer computation (VT-Aligned) yields an additional **+2.5%** improvement at zero cost—no model re-inference, no new data. Combined with the V7.1 adjoint state prefetch method, the stacked **VT-V7** configuration achieves **+29.6%** over the cosine baseline on NFCorpus (NDCG@10 = 0.2844), setting a new all-time best for the Shape-CFD framework. A native Rust implementation reduces end-to-end latency to **38 ms** per query.

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1 Introduction

Retrieval-Augmented Generation (RAG) systems have emerged as the dominant architecture for grounding large language models in external knowledge [10, 11]. In a typical RAG pipeline, a bi-encoder first retrieves a set of N candidate passages from a large corpus using approximate nearest-neighbor (ANN) search, and then either these candidates are fed directly to the generator or an intermediate *reranker* adjusts the ordering before generation. The quality of the final answer critically depends on the precision and recall of this retrieval stage.

The prevailing approach to first-stage ranking relies on cosine similarity between the query vector q and each document vector v_i, computed independently: $$\text{score}(d_i) = \frac{q \cdot v_i}{\|q\| \|v_i\|}$$ While simple and efficient, this pointwise scoring paradigm fundamentally ignores the **inter-document semantic structure** within the retrieved set. When the query is ambiguous, polysemous, or spans multiple topical domains, isolated scoring fails to exploit the mutual information carried by document neighborhoods—a limitation well-studied under the *cluster hypothesis* in information retrieval [4].

Cross-encoder rerankers [11] address this quality gap by jointly encoding the query-document pair through a deep transformer, achieving state-of-the-art ranking accuracy. However, their computational cost scales as $O(N \cdot L^2 \cdot H)$, where L is the sequence length and H the hidden dimension, resulting in latencies of approximately 500 ms for $N = 30$ candidates—prohibitive for many real-time applications.

Recent work has explored graph-based diffusion as a middle ground. Dampanaboina et al. [2] construct a document similarity graph and apply isotropic heat diffusion to re-score candidates, demonstrating that inter-document signal propagation can improve retrieval quality without the overhead of cross-encoders. However, their formulation is purely *diffusive*: every node diffuses its score to neighbors symmetrically, without directional preference. In physical terms, this corresponds to a system governed solely by Fick’s second law, which guarantees convergence to a flat equilibrium—the retrieval analogue of over-smoothing in graph neural networks [13, 14].

Our key insight is that retrieval reranking is more faithfully modeled by the full *convection-diffusion equation* rather than pure diffusion. In fluid dynamics, the convection-diffusion equation $\frac{\partial C}{\partial t} = D\nabla^2 C - \nabla \cdot (\vec{u}C)$ governs the transport of a scalar concentration field C under the combined influence of diffusive spreading (coefficient D) and directional advective transport (velocity field $\vec{u}$). By drawing this analogy, we introduce an **advection term** that encodes the query’s semantic direction: on each graph edge, the advection coefficient measures the alignment between the inter-document displacement vector and the query embedding, providing a directional bias that pure diffusion lacks.

A second limitation of existing approaches is that they represent each document as a single point in the embedding space. In practice, a document—particularly a statutory provision or a technical passage—carries heterogeneous semantic content across its constituent sentences. A single centroid embedding necessarily conflates these distinct semantic facets. To address this, we

propose the **Shape-CFD extension**, which represents each document as a *sentence-level point cloud* $\mathcal{P}_i = \{s_{i,1}, \ldots, s_{i,K_i}\} \subset \mathbb{R}^d$ and replaces cosine-based graph weights with the symmetric Chamfer distance. This formulation captures local semantic matches between document sub-components—even when the overall document-level similarity is moderate—and naturally degrades to the point representation for single-sentence documents.

However, computing Chamfer distances in the native $d = 4096$ embedding space introduces a dual challenge: (i) the computational cost of $O(K_A \cdot K_B \cdot d)$ per document pair, and (ii) the *curse of dimensionality*, which causes inner-product-based advection coefficients to collapse toward zero (with standard deviation $\sigma \approx 1/\sqrt{d} \approx 0.016$). We resolve both issues via Johnson-Lindenstrauss (JL) random projection to $m = 128$ dimensions, achieving a $32\times$ speedup. Crucially, we provide a corrected theoretical account of the dimensional reduction effect: contrary to naive interpretations, the expanded variance of inner products in the reduced space does not "amplify" true physical signals but rather injects $O(1/m)$ stochastic noise that acts as a form of *flow-field annealing*, breaking the near-orthogonal deadlock of high-dimensional geometry.

Contributions

We summarize our contributions as follows:

1. **AD-Rank**: A graph-based reranking algorithm that solves the conservative convection-diffusion equation on a sparse KNN document graph, introducing directional query-aware advection to complement isotropic diffusion. We prove global mass conservation and convergence under CFL stability conditions.

2. **Shape-CFD**: An extension that replaces point embeddings with sentence-level point clouds, using Chamfer distance for graph construction and centroid-based advection directions. This captures fine-grained sub-document semantic matching.

3. **Theoretical analysis**: We characterize the operating regime through the structural Péclet number and provide a physically grounded explanation for the empirically observed optimality of weak advection ($\text{Pe} \approx 0.065$).

4. **Comprehensive evaluation**: We evaluate on three BEIR benchmarks (SciFact, NFCorpus, FiQA) and a Chinese legal statute retrieval dataset (23,701 documents), demonstrating competitive NDCG@10 with 22 ms latency on CPU.

The remainder of this paper is organized as follows. Section 2 surveys related work. Section 3 presents the mathematical framework of AD-Rank and its Shape-CFD extension. Section 4 reports experimental results. Section 5 provides physical analysis of the algorithm's behavior, and Section 6 concludes with directions for future work.

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2 Related Work

Our work draws upon and bridges three distinct research threads: graph-based reranking in information retrieval, distributional and shape-aware representations for text, and partial differential equations on graph structures.

2.1 Graph-Based Reranking

The idea of leveraging inter-document relationships for reranking has a long history. The PageRank algorithm [4] pioneered the use of iterative graph propagation for web page ranking, treating hyperlink structure as a random walk on a directed graph. More recently, several works have adapted graph-based methods specifically for dense retrieval reranking.

Diffusion-Aided RAG [2] constructs a semantic similarity graph over retrieved documents and applies isotropic heat diffusion to redistribute relevance scores. By allowing high-scoring documents to propagate their scores to semantically similar neighbors, this approach effectively discovers latent relevant passages that might have received low initial cosine scores. However, the diffusion process is purely isotropic—it lacks any directional preference tied to the query—and therefore cannot distinguish between semantically similar documents that differ in their query-relevance orientation.

G-RAG [18] employs graph neural networks to learn reranking functions over document graphs, achieving strong results through supervised training on labeled query-document relevance pairs. While powerful when sufficient training data is available, this supervised approach requires domain-specific annotation and does not transfer freely across domains without fine-tuning.

Graph-Augmented Dense Statute Retrieval (G-DSR) [19] exploits the hierarchical structure inherent in legal document collections, using graph augmentation to improve statute retrieval. Our approach differs fundamentally in that we construct graphs dynamically from the retrieved set at query time, making no assumptions about pre-existing document structure.

Spreading Activation for KG-RAG [20] adapts the classical spreading activation paradigm from cognitive science to knowledge graph-augmented retrieval. Activation propagation can be viewed as a special case of diffusion; our framework generalizes this by introducing the advection term for directional transport.

2.2 Shape and Distributional Representations

The standard practice in dense retrieval compresses each document into a single vector (a “point” in embedding space), which necessarily conflates the diverse semantic facets of multi-sentence passages. Several prior works have explored richer representations.

Gaussian Embeddings [1] represent words as Gaussian distributions rather than point vectors, capturing uncertainty and enabling asymmetric distance measures via KL divergence. The GELD framework [21] extends this to document-level Gaussian embeddings for retrieval.

ColBERT [22] introduces a *late interaction* mechanism in which both query and document are represented as sets of token-level vectors, and relevance is computed via a *MaxSim* operator that finds the maximum similarity between each query token and all document tokens. This multi-vector representation captures fine-grained alignment but computes relevance in a purely pointwise manner—each document is scored independently of its retrieved neighbors.

Our Shape-CFD extension shares ColBERT’s insight that documents should be represented as collections of sub-vectors. However, we go further by using these point clouds not for isolated pointwise scoring but as the basis for *graph-structured collective reranking*: the Chamfer distance between point clouds determines graph edge weights, and the full convection-diffusion dynamics then propagates information across the document graph. This listwise, topology-aware approach is fundamentally distinct from ColBERT’s pointwise MaxSim.

2.3 Partial Differential Equations on Graphs

The application of continuous PDEs to discrete graph structures has gained significant traction in the graph neural network (GNN) community.

GRAND [13] established that many GNN architectures can be interpreted as discretizations of the heat diffusion equation on graphs, providing a continuous-time perspective on message passing. This insight has spawned a family of diffusion-based GNN architectures.

Reaction-Diffusion GNNs [23] extend the pure diffusion framework by incorporating nonlinear reaction terms, enabling the formation of *Turing patterns*—spatially heterogeneous steady states that counteract the over-smoothing tendency of pure diffusion. The Allen-Cahn reaction term, in particular, drives a phase-separation dynamic that polarizes node features into distinct clusters. Our analysis of the Péclet number regime (Section 5) reveals a related phenomenon: the dominant diffusion term provides global smoothing while the weak advection term introduces a subtle but critical directional perturbation.

Advection-Augmented CNNs [3] directly incorporate advection-diffusion dynamics into convolutional neural network layers for scientific prediction tasks, demonstrating that the advective transport component meaningfully enhances model performance. Our work is, to our knowledge, the first to apply the full convection-diffusion equation to information retrieval reranking.

Spectral methods on graphs, including Chebyshev spectral filtering [24] and diffusion kernels via the matrix exponential $K = \exp(-\beta L)$ [12], provide alternative computational pathways. For our problem scale ($N = 30$ candidate documents), explicit time-stepping proves more practical than eigendecomposition, though spectral acceleration remains a promising direction for larger candidate sets.

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3 Method

We now present the AD-Rank algorithm and its Shape-CFD extension. We begin by formalizing the reranking problem, then describe the graph construction, the conservative convection-diffusion discretization, convergence guarantees, and the shape-aware generalization.

3.1 Problem Formulation

Let $q \in \mathbb{R}^d$ denote the query embedding produced by a pretrained language model (in our implementation, Qwen3-Embedding-8B with $d = 4096$). Given a corpus, the first-stage retriever returns a candidate set $\mathcal{D} = \{d_1, d_2, \ldots, d_N\}$ of N documents (we use $N = 30$ via HNSW approximate nearest-neighbor search), where each document d_i is associated with an embedding vector $v_i \in \mathbb{R}^d$.

The reranking task is to learn a scoring function $f: \mathcal{D} \times q \to \mathbb{R}^N$ that produces a ranking superior to the initial cosine-similarity ordering. Unlike pointwise and pairwise rerankers, our approach is *listwise*: the score of each document is determined collectively by the entire retrieved set through a dynamic process on a document graph.

3.2 Graph Construction

We construct a sparse, weighted, undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, W)$ over the N candidate documents.

Step 1: Pairwise similarity. We compute the full $N \times N$ cosine similarity matrix: $$S_{ij} = \cos(v_i, v_j) = \frac{v_i \cdot v_j}{\|v_i\| \|v_j\|}, \quad \forall i,j \in \{1, \ldots, N\}$$

Step 2: KNN sparsification. For each node i, we retain edges only to its k most similar neighbors: $$\mathcal{N}(i) = \text{argTopK}_{j \neq i}(S_{ij}, k), \quad k = 3$$ This sparsification reduces the graph density from $O(N^2)$ to $O(Nk)$, which is critical for controlling both computational cost and diffusion dynamics.

Step 3: Edge weights. We assign exponentially decaying weights: $$W_{ij} = \exp\left(-\beta \cdot (1 - S_{ij})\right), \quad \beta = 2.0$$ The parameter β controls the decay rate: larger values concentrate diffusion along high-similarity edges.

Step 4: Symmetrization. If $j \in \mathcal{N}(i)$ but $i \notin \mathcal{N}(j)$, we add the reverse edge with $W_{ji} = W_{ij}$. This ensures the graph is undirected, which is essential for the symmetric positive-definite property of the graph Laplacian and, consequently, for the conservation law of the diffusion operator.

3.3 Conservative Convection-Diffusion on Graphs

We model the evolution of document relevance scores as a convection-diffusion process on the graph $\mathcal{G}$.

Initial concentration field. Each document is assigned an initial "concentration" equal to its cosine similarity with the query: $$C_0(i) = \cos(q, v_i), \quad i = 1, \ldots, N$$ This initial condition seeds the dynamic process with the first-stage retrieval signal.

Advection direction. For each edge (i, j), we define the advection coefficient by projecting the inter-document displacement onto the query direction: $$u_{ij} = \left\langle \frac{v_j - v_i}{\|v_j - v_i\|_2 + \varepsilon},\ \frac{q}{\|q\|} \right\rangle \cdot u_{\text{strength}}, \quad \varepsilon = 10^{-8}$$

The scalar u_{ij} measures how well the edge direction $i \to j$ aligns with the query's semantic orientation. A positive value ($u_{ij} > 0$) indicates that moving from i to j proceeds "downwind" (toward the query); a negative value indicates the reverse. The small constant ε prevents division by zero when two documents have near-identical embeddings.

Anti-symmetry. The advection coefficients satisfy a strict anti-symmetry: $$u_{ji} = \left\langle \frac{v_i - v_j}{\|v_i - v_j\|},\ \hat{q} \right\rangle \cdot u_{\text{strength}} = -u_{ij}$$ since $(v_i - v_j = -(v_j - v_i)$ and $\|v_i - v_j\| = \|v_j - v_i\|$. This property is crucial for ensuring global conservation of the advective flux ($\sum_i F_i^{\text{adv}} = 0$).

Governing equation. The continuous convection-diffusion equation $$\frac{\partial C}{\partial t} = D \nabla^2 C - \nabla \cdot (\vec{u} C)$$ is discretized on the graph using a **conservative upwind scheme**:

$$C_{t+1}(i) = C_t(i) + \Delta t \cdot \Bigg[\underbrace{D \sum_{j \in \mathcal{N}(i)} W_{ij}\, \big(C_t(j) - C_t(i)\big)}_{\text{diffusion flux}} + \underbrace{\sum_{j \in \mathcal{N}(i)} W_{ij} \Big[\max(u_{ji}, 0)\, C_t(j) - \max(u_{ij}, 0)\, C_t(i)\Big]}_{\text{conservative upwind advection flux}} \Bigg]$$

The upwind discretization is essential for numerical stability. A naive central-difference scheme ($u_{ij} W_{ij} (C_j - C_i)$) produces non-physical oscillations when the local Péclet number

exceeds 2. The upwind scheme resolves this by always sampling concentration from the upstream node, preserving the causal direction of information transport.

Global conservation. We prove that the total concentration is conserved at each time step:

$$\sum_{i=1}^{N} F_i^{\text{adv}} = \sum_{i=1}^{N} \sum_{j \in \mathcal{N}(i)} W_{ij} \left[\max(u_{ji}, 0)\, C_j - \max(u_{ij}, 0)\, C_i\right] = 0$$

Proof. Since the graph is undirected ($W_{ij} = W_{ji}$) and $u_{ji} = -u_{ij}$, each ordered pair (i, j) contributes a flux of $W_{ij} \max(u_{ji}, 0) C_j$ to node i and $W_{ji} \max(u_{ij}, 0) C_i$ outward from node j (equivalently, $-W_{ij} \max(u_{ij}, 0) C_i$ at node i). Summing over all edges, each pair's contributions cancel exactly. $\blacksquare$

This conservation property guarantees that the reranking process redistributes relevance scores without creating or destroying total "score mass"—a desirable property that prevents score inflation or deflation.

CFL stability condition. To ensure absolute stability of the explicit time integration (non-negative concentrations and absence of numerical oscillations), we require that the coefficient of $C_t(i)$ in the update equation remains non-negative: $$1 - \Delta t \sum_{j \in \mathcal{N}(i)} W_{ij} \left[D + \max(u_{ij}, 0)\right] \geq 0$$

This yields the strict CFL time-step bound: $$\Delta t_{\text{CFL}} = \frac{1}{\max_{i} \sum_{j \in \mathcal{N}(i)} W_{ij} \left(D + \max(u_{ij}, 0)\right)}$$

In practice, we use $\Delta t = C_{\text{safe}} \cdot \Delta t_{\text{CFL}}$ with safety factor $C_{\text{safe}} = 0.9$. The iteration terminates when $\max_i |C_{t+1}(i) - C_t(i)| < \epsilon$ with $\epsilon = 10^{-3}$, which typically requires 30–50 steps.

3.4 Convergence Analysis

Theorem 1 (Global Convergence). *Given a connected graph $\mathcal{G}$ and a time step Δt satisfying the CFL condition of Section 3.3, the iterative sequence $\{C_t\}$ generated by the conservative upwind scheme converges to a unique stationary distribution C^*, and the total concentration $\sum_i C_t(i)$ is conserved for all t.*

Proof. We write the update in matrix form $C_{t+1} = \mathbf{M}\, C_t$, where the transition matrix $\mathbf{M}$ has entries: - Diagonal: $\mathbf{M}_{ii} = 1 - \Delta t \sum_{k \in \mathcal{N}(i)} W_{ik} \left[D + \max(u_{ik}, 0)\right]$ - Off-diagonal ($j \in \mathcal{N}(i)$): $\mathbf{M}_{ij} = \Delta t \cdot W_{ij} \left[D + \max(u_{ji}, 0)\right]$

We establish the four conditions for the Perron-Frobenius theorem:

(i) Non-negativity. The CFL condition ensures $\mathbf{M}_{ii} \geq 0$; all off-diagonal entries are non-negative since $D, W_{ij} \geq 0$.

(ii) Column-stochasticity. By the symmetry of edge weights ($W_{ij} = W_{ji}$) and the structure of the upwind scheme, the column sums satisfy $\sum_i \mathbf{M}_{ij} = 1$ for all j. Thus $\mathbf{M}$ is column-stochastic, and the total concentration is conserved.

(iii) Irreducibility. When the graph $\mathcal{G}$ is connected, $\mathbf{M}$ is irreducible.

(iv) Aperiodicity. Since $\mathbf{M}_{ii} > 0$ for all i (the CFL bound is strict), every node has a self-loop, guaranteeing aperiodicity.

By the Perron-Frobenius theorem, the dominant eigenvalue $\lambda_1 = 1$ is simple, and all other eigenvalues satisfy $|\lambda_k| < 1$. Therefore, $C_t = \mathbf{M}^t C_0$ converges exponentially to the unique stationary distribution C^*. $\blacksquare$

Remark on graph connectivity. In practice, a KNN graph with $N = 30$ and $k = 3$ may decompose into multiple connected components, particularly for cross-domain queries that fragment the candidate set into semantic islands. When the graph has $K > 1$ connected components, the eigenvalue $\lambda_1 = 1$ has algebraic multiplicity K, and the stationary distribution is no longer unique—it depends on the initial allocation of concentration mass across components. We note that this is, in fact, a desirable property: it naturally prevents *score leakage* between unrelated semantic clusters. In our implementation, we do not enforce artificial connectivity (e.g., via teleportation as in PageRank), allowing each semantic sub-manifold to evolve independently.

Early stopping and transient dynamics. The fully converged stationary C^* depends only on the graph topology and advection field, and is independent of the initial condition C_0. In practice, we terminate at $\epsilon = 10^{-3}$ (typically 30–50 iterations), capturing the *transient state* where the initial retrieval signal (C_0) has been partially—but not completely—smoothed by the diffusion dynamics. This transient truncation is not merely an engineering convenience but a deliberate design choice: it preserves the first-stage cosine relevance signal while enriching rankings with inter-document contextual information.

3.5 Adaptive Parameters

Fiedler-value adaptive diffusion. The algebraic connectivity of the graph (the second-smallest eigenvalue λ_2 of the graph Laplacian $L = D_{} - W, known as the Fiedler value) provides a natural measure of graph density. We adapt the diffusion coefficient accordingly: $$D_{\text{adaptive}} = \frac{D_{\text{base}}}{1 + \lambda_2}$$ When λ_2 is large (tightly clustered candidates), D is reduced to prevent over-smoothing; when λ_2 is small (sparse, dispersed candidates), D is increased to promote signal propagation.

Cross-domain adaptive advection. The variance of inter-document cosine similarities σ^2_{sim} serves as a proxy for semantic heterogeneity. We adjust the advection strength via linear interpolation: $$u_{\text{adaptive}} = u_{\min} + (u_{\max} - u_{\min}) \cdot \min\left(\frac{\sigma^2_{\text{sim}}}{0.04},\ 1\right)$$ Higher variance (indicating cross-domain queries) triggers stronger advection to exploit the more pronounced directional gradients.

3.6 Shape-CFD Extension

The core innovation of Shape-CFD is the elevation of document representations from single points to multi-point *shapes* (point clouds).

Point cloud representation. Each document d_i is decomposed into its constituent sentences, and each sentence is independently embedded to form a point cloud: $$d_i \to \mathcal{P}_i = \{s_{i,1}, s_{i,2}, \ldots, s_{i,K_i}\} \subset \mathbb{R}^d$$ where K_i is the number of sentences in document d_i. Single-sentence documents naturally reduce to singleton point clouds, ensuring backward compatibility with the point-based AD-Rank.

Chamfer distance as Unbalanced Optimal Transport. We measure inter-document affinity using the symmetric Chamfer distance: $$d_{\text{Chamfer}}(\mathcal{P}_A, \mathcal{P}_B) = \frac{1}{|\mathcal{P}_A|} \sum_{a \in \mathcal{P}_A} \min_{b \in \mathcal{P}_B} d_{\cos}(a, b) + \frac{1}{|\mathcal{P}_B|} \sum_{b \in \mathcal{P}_B} \min_{a \in \mathcal{P}_A} d_{\cos}(a, b)$$ where $d_{\cos}(a, b) = 1 - \cos(a, b)$ is the cosine distance. Unlike the centroid cosine similarity, Chamfer captures *local* semantic matches: even when two documents are

globally dissimilar, high similarity between specific sentence pairs yields a small Chamfer distance.

Crucially, Chamfer distance is not an approximation to the Wasserstein-1 (Earth Mover's) distance, but rather a fundamentally different object. The standard W_1 distance enforces strict mass conservation via marginal constraints: every unit of semantic mass in the source must be transported to a unique location in the target. For retrieval, this creates a catastrophic *noise penalty*: when a query contains 1 sentence and a document contains 50 sentences (only 1 relevant), W_1 forces the 49 irrelevant sentences to be matched somewhere, generating spurious transport cost that overwhelms the genuine relevance signal.

Chamfer distance operates as an *Unbalanced Optimal Transport* (UOT) without marginal constraints. It allows many-to-one mappings and abandons unmatched points entirely, functioning as a parameter-free hard semantic attention mechanism that isolates the most relevant sentence pairs while discarding noise. This property makes Chamfer distance particularly well-suited for long, heterogeneous documents—precisely where Shape-CFD provides the greatest advantage (NFCorpus: +24.6% vs. cosine).

Shape-aware graph weights and initial conditions. We replace the cosine-based graph construction with Chamfer-based equivalents: $$W_{ij}^{\text{shape}} = \exp\left(-\beta \cdot d_{\text{Chamfer}}(\mathcal{P}_i, \mathcal{P}_j)\right)$$ $$C_0(i) = \exp\left(-2 \cdot d_{\text{Chamfer}}(\mathcal{P}_q, \mathcal{P}_i)\right)$$

The exponential mapping ensures non-negative initial concentrations ($C_0 \geq 0$), resolving the physical inconsistency of negative concentrations that arises when using raw cosine similarities (which can be negative) as initial conditions in the diffusion process.

Centroid-based advection. For the advection direction, we compute the normalized centroid of each point cloud: $$\bar{v}_i = \frac{\sum_{k=1}^{K_i} s_{i,k}}{\left\|\sum_{k=1}^{K_i} s_{i,k}\right\|_2}$$ and define the shape-aware advection coefficient as: $$u_{ij}^{\text{shape}} = \left\langle \frac{\bar{v}_j - \bar{v}_i}{\|\bar{v}_j - \bar{v}_i\|},\ \frac{\bar{v}_q}{\|\bar{v}_q\|} \right\rangle \cdot u_{\text{strength}}$$

The explicit L_2-normalization of the centroid sum is critical: without it, documents with many semantically diverse sentences would have centroids that collapse toward the origin ($\|\bar{v}_i\| \to 0$), introducing a severe document-length bias into the advection field.

Metric space inconsistency (known limitation). We acknowledge a fundamental tension in this formulation: the graph topology is constructed using Chamfer distance (a set-theoretic metric on point clouds), while the advection coefficients are computed from centroid differences (a Euclidean metric on single points). This places the diffusion and advection operators in two different metric spaces—a gap that is theoretically unsatisfying. We adopt this hybrid approach as a pragmatic engineering compromise in the current version: Chamfer distance captures fine-grained inter-document structure, while centroid-based advection remains computationally tractable. In Section 6, we propose *scalar potential convection* as a principled resolution that eliminates this inconsistency entirely.

Noise dilution in symmetric Chamfer (known limitation). The symmetric Chamfer distance averages over *both* forward and reverse nearest-neighbor distances. In retrieval scenarios where a long document ($K_i = 50$ sentences) contains only a single highly relevant sentence, the remaining 49 unmatched sentences contribute large cosine distances to the reverse term, diluting the signal from the one critical match. This *noise dilution* effect systemically penalizes long, heterogeneous documents—precisely the documents where shape-aware representations should provide the greatest benefit. Asymmetric alternatives, such as the directed Hausdorff distance or ColBERT-style MaxSim, would address this issue at the cost of losing the metric symmetry

required by our undirected graph framework. We defer the formal resolution to future work (Section 6).

3.7 Acceleration via Johnson-Lindenstrauss Projection

Computing Chamfer distances in the native $d = 4096$ space has complexity $O(K_A \cdot K_B \cdot d)$ per document pair. For $N = 30$ candidates with an average of $K = 5$ sentences each, this amounts to $\binom{30}{2} \times 25 \times 4096 \approx 4.5 \times 10^7$ multiply-adds—feasible but expensive for real-time applications.

We apply a random projection $P \in \mathbb{R}^{m \times d}$ with $P_{ij} \sim \mathcal{N}(0, 1/m)$ and $m = 128$: $$\tilde{s} = \frac{P \cdot s}{\|P \cdot s\|_2}$$

The post-projection L_2-normalization ensures that the equivalence $\|x - y\|_2^2 = 2(1 - \cos(x, y))$ holds for unit vectors, allowing the JL distance-preservation guarantee to transfer to the cosine metric.

Johnson-Lindenstrauss Lemma [7]. For any $\epsilon > 0$ and point set of size n, when $m \geq O(\epsilon^{-2} \log n)$: $$(1 - \epsilon)\|s_a - s_b\|^2 \leq \|\tilde{s}_a - \tilde{s}_b\|^2 \leq (1 + \epsilon)\|s_a - s_b\|^2$$

Dimensional reduction effects and their limitations. In the reduced $m = 128$ space, the standard deviation of random inner products increases from $\sigma = 1/\sqrt{4096} \approx 0.016$ to $\sigma = 1/\sqrt{128} \approx 0.088$—a factor of $\sim 5.6\times$. The JL lemma guarantees that the *expected value* of pairwise distances is preserved ($\mathbb{E}[\|\tilde{s}_a - \tilde{s}_b\|^2] = \|s_a - s_b\|^2$), so the graph topology constructed from Chamfer distances remains faithful in expectation. However, the expanded variance of inner products in the reduced space is *projection noise*, not signal amplification. This noise introduces $O(1/m)$ stochastic perturbations into the advection coefficients.

We make no claim that this noise is physically beneficial. The advection coefficients in the native $d = 4096$ space are inherently weak due to the concentration of measure on the unit sphere ($\sigma \approx 0.016$), and the JL projection does not resolve this fundamental limitation—it merely trades the curse of dimensionality for projection variance. The empirical effectiveness of the algorithm in the projected space should be attributed to the *diffusion-dominated regime* (Pe ≈ 0.065, where advection plays a secondary role) rather than to any purported “signal amplification” of the projection. As discussed in Section 6, scalar potential convection—which operates on scalar score differences rather than vector inner products—offers a principled alternative that is immune to the curse of dimensionality.

Computational speedup. The primary justification for JL projection is computational: the dimensionality reduction yields a $d/m = 32\times$ speedup in all distance computations. Combined with offline precomputation and SQLite caching of projected sentence vectors (a one-time cost), the end-to-end Shape-CFD latency is reduced from 1,837 ms (with live API embedding) to **22 ms**.

3.8 Dual Physical Indicators

We introduce two complementary dimensionless numbers to characterize the algorithm’s operating regime:

Dynamic Reynolds number (runtime indicator): $$\text{Re} = \frac{\sum_{i=1}^{N} |F_i^{\text{adv}}|}{\sum_{i=1}^{N} |F_i^{\text{diff}}|}$$ This measures the ratio of advective

to diffusive *activity* (taking absolute values at each node to avoid the trivial cancellation inherent in symmetric graphs).

Structural Péclet number (static indicator): $$\text{Pe} = \frac{\overline{|u_{ij}|}}{D} = \frac{\frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} |u_{ij}|}{D}$$ This is computed once at graph construction time, independent of the concentration field C, and captures the intrinsic balance between advective and diffusive capacity of the graph structure.

---

4 Experiments

4.1 Datasets

We evaluate Shape-CFD on four datasets spanning two languages and three domains:

BEIR Benchmarks. We use three subsets from the BEIR benchmark suite [25], widely adopted for evaluating zero-shot retrieval:

Dataset	Domain	Queries	Corpus Size	Avg. Doc Length
SciFact	Scientific claims	300	5,183	Short abstracts
NFCorpus	Biomedical	323	3,633	Mixed lengths
FiQA	Financial QA	648	57,638	Variable

LawVein Legal Corpus. A proprietary Chinese legal knowledge base consisting of 23,701 statute chunks extracted from 24 Chinese laws, encoded with Qwen3-Embedding-8B ($d = 4096$) and indexed using USearch HNSW ($M = 32$), ($\text{ef} = 128$). This dataset serves as our primary development testbed and provides a challenging scenario with highly homogeneous intra-domain embeddings.

4.2 Baselines and Evaluation Protocol

Retrieval methods compared:

- **Cosine**: Direct HNSW retrieval with cosine similarity ranking (the standard baseline).
- **BM25**: Sparse lexical matching via Okapi BM25.
- **AD-Rank v2**: Our point-based convection-diffusion algorithm (without the shape extension), using optimal parameters $D = 0.15$, $u_{\text{strength}} = 0.1$, $k = 3$.
- **Shape-CFD**: The full shape-aware extension with Chamfer distance, JL projection ($m = 128$), and sentence-level point clouds.

Strategy variants (for the LawVein ablation study):

Variant	Description	Key Parameter
MDA	Multi-Directional Advection: 8 random projection subspaces, each with independent graph and advection	$M = 8$, $k = 32$
BAA	Block-Adaptive Advection: 8 contiguous dimension blocks	$B = 8$
v2+PCA	Local PCA dimensionality reduction for advection direction	$\text{pcaDim} = 32$

Variant	Description	Key Parameter
v2+Temperature	Nonlinear temperature scaling of advection coefficients	$\alpha = 0.5$
v2+Adaptive	Fiedler-value adaptive D + cross-domain adaptive u	Auto

Evaluation metrics. For BEIR benchmarks, we report NDCG@10, MRR@10, and Recall@10 using the official evaluation scripts. For the LawVein dataset (which lacks ground-truth relevance labels), we employ **DeepSeek V3.2 blind evaluation**: an LLM judge independently scores anonymized A/B result pairs on a 1–5 scale for relevance, coverage, and ranking rationality, with random assignment to prevent position bias.

4.3 Main Results

BEIR Benchmark Results

Dataset	Cosine	BM25	AD-Rank v2	Shape-CFD	Δ NDCG@10
SciFact	0.4701	—	0.4753	**0.4820**	**+2.5%**
NFCorpus	0.2209	—	0.2398	**0.2684**	**+21.5%**
FiQA	0.1683	—	0.1683	0.1683	0%

On SciFact and NFCorpus, Shape-CFD substantially outperforms both the cosine baseline and the point-based AD-Rank v2. The NFCorpus improvement of +21.5% is particularly notable, as NFCorpus contains biomedical documents with rich multi-sentence structure that benefits from the Chamfer distance-based graph construction.

The FiQA result ($\Delta = 0$) is informative: FiQA's financial Q&A documents tend to be short, single-sentence answers that yield singleton point clouds. In this regime, Shape-CFD degrades exactly to cosine similarity (as expected), confirming that the shape extension contributes only when multi-sentence structure is present—it does not introduce spurious effects on incompatible document types.

LawVein Blind Evaluation Results

AD-Rank v2 vs. Cosine (10 queries, DeepSeek V3.2 blind judge):

Method	Mean Score	Win	Loss	Draw	Win Rate
AD-Rank v2	3.4/5	7	2	1	**70%**
Cosine	2.8/5	2	7	1	20%

Shape-CFD vs. Cosine (10 queries, independent evaluation):

Method	Mean Score	Win	Loss	Win Rate
Shape-CFD	3.4/5	**8**	2	**80%**
Cosine	2.6/5	2	8	20%

Shape-CFD achieves an 80% win rate against cosine (8:2), surpassing AD-Rank v2's 70% win rate (7:2:1). The improvement is most pronounced on cross-domain queries (e.g., "divorce property division" spanning civil law and procedural law), where the directional advection and multi-sentence matching provide the greatest benefit.

Full Strategy Cross-Evaluation (LawVein)

To validate AD-Rank v2 as the optimal point-based configuration, we conducted a comprehensive 7-method tournament with $10 \times 6 = 60$ blind comparisons:

Rank	Method	Blind Score	Pe	Latency	Convergence	W/L
🥇	**AD-Rank v2**	**3.4/5**	0.065	14.8 ms	**100%**	**7/2**
🥈	BAA	3.3/5	0.226	7 ms	20%	7/3
🥉	v2+PCA	3.2/5	0.112	33 ms	90%	6/4
4	v2+Adaptive	3.0/5	0.122	3.3 ms	100%	5/4
5	v2+Temperature	2.9/5	0.604	3.5 ms	10%	5/5
6	MDA	2.7/5	0.351	85.8 ms	0%	3/7

Key findings: (1) AD-Rank v2 achieves the highest blind evaluation score with 100% convergence. (2) Higher Péclet numbers do *not* correlate with better ranking quality—the methods with the strongest advection (MDA: Pe=0.351, Temperature: Pe=0.604) perform *worst*, supporting our hypothesis that weak advection is optimal. (3) Convergence rate is a hard constraint: methods with convergence rates below 50% (BAA, Temperature, MDA) produce inconsistent results.

4.3.3 V5 Reaction-Diffusion Ablation (v3 supplement)

In Section 6 of v1, we proposed augmenting the convection-diffusion equation with an Allen-Cahn bistable reaction term $R(C) = \gamma\, C(1-C)(C-\theta)$ to produce genuinely heterogeneous stationary distributions. We report systematic experiments testing this proposal. All experiments in Sections 4.3.3–4.3.4 use a unified evaluation framework operating on the native $d = 4096$ embedding space (without JL projection), with the Chamfer-distance graph topology and $N = 30$, $k = 3$.

V5.0: Unnormalized Allen-Cahn. We introduce the reaction term with $\gamma = 0.5$ and threshold θ set to the mean of the initial concentration field. On NFCorpus, this yields NDCG@10 = 0.2467 (-1.6% vs. Shape-CFD v4 baseline of 0.2508), confirming that the naive addition of reaction dynamics degrades performance. The root cause: when $C_0 \in [0.1, 0.2]$ (a typical range for cosine-transformed scores), the reaction term evaluates to $R(C) \approx 0.004$, which after multiplication by $\Delta t = 0.05$ contributes only ~ 0.0002 per step—completely overwhelmed by the diffusion flux.

V5.1: Normalized Allen-Cahn. Following diagnostic analysis, we attempted to amplify the reaction dynamics by applying Min-Max normalization to map C_0 to $[0,1]$ before applying $R(C)$ with $\gamma = 5.0$. This produced a catastrophic -12.9% regression on NFCorpus (NDCG@10 = 0.2184). The failure mechanism: Min-Max normalization stretches a natural range of $\Delta \approx 0.1$ to the full $[0,1]$ interval—an effective spatial dilation of $10\times$. The combination with $\gamma = 5.0$ yields an effective reaction rate of $\gamma_{\text{eff}} \approx 500$ in the original coordinate system, triggering violent phase separation that destroys the carefully calibrated ranking signal. This result empirically confirms that γ must be scaled inversely with the square of the score range.

V5.2: Domain-adapted Allen-Cahn. We implement the corrected formulation: $$\gamma_{\text{eff}} = \frac{\gamma_{\text{base}}}{(C_{\max} - C_{\min} + \epsilon)^2}, \quad R(C) = \gamma_{\text{eff}} (C - C_{\min})(C_{\max} - C)(C - \theta_{\text{Otsu}})$$

where θ_{Otsu} is obtained via Otsu's method on the initial concentration histogram. We additionally implement Zelnik-Manor local bandwidth scaling: $W_{ij} = W_{ij}^0 \cdot \exp(-\|v_i - v_j\|^2 / (\sigma_i \sigma_j))$, where σ_i is the distance from node i to its k-th nearest neighbor.

Configuration	NFCorpus NDCG@10	Δ vs V4
Shape-CFD v4 (baseline)	0.2508	—
V5.0: $\gamma=0.5$, unnormalized	0.2467	-1.6%
V5.1: $\gamma=5.0$, normalized	0.2184	-12.9%
V5.2: $\gamma_{\text{base}}=0.5$, domain-adapted	0.2501	-0.3%
V5.2: $\gamma_{\text{base}}=1.0$, domain-adapted	0.2501	-0.3%
V5.2: $\gamma_{\text{base}}=2.0$, domain-adapted	0.2501	-0.3%

The domain-adapted formulation (V5.2) successfully prevents the catastrophic phase transition of V5.1, recovering to within 0.3% of the baseline. However, three values of γ_{base} (0.5, 1.0, 2.0) produce *identical* results (0.2501), demonstrating that the reaction term's contribution is negligible—the system is **topologically locked** to a Dirichlet energy minimum where the small-sample KNN graph ($N=30$, diameter ≤ 4) allows diffusion to equilibrate instantaneously, rendering any perturbation from the reaction term invisible to the NDCG ranking metric.

Conclusion. In the short-query, narrow score-range regime ($\Delta C_0 \approx 0.1$), the Allen-Cahn reaction term is *numerically dormant*—not due to implementation error, but as a physical necessity. The reaction force scales as $O(\Delta^3)$ while diffusion scales as $O(\Delta)$; when $\Delta \ll 1$, diffusion dominates by construction. This result validates the v1 observation (Section 5.1) that the optimal operating regime is diffusion-dominated, and precisely delineates the boundary conditions under which nonlinear dynamics become physically meaningful: *only when the initial score range is sufficiently large* ($\Delta C_0 \gg 0.1$), as expected in long-document retrieval scenarios with sparse evidence.

4.3.4 Mixed Initial Field Ablation (v3 supplement)

The v1 Shape-CFD formulation uses the symmetric Chamfer distance to compute both graph weights and initial concentrations. For single-vector queries, the query-document Chamfer distance $d_{\text{Chamfer}}(q, \mathcal{P}_i)$ decomposes into two components:

$$d_{\text{Chamfer}}(q, \mathcal{P}_i) = \underbrace{\min_{s \in \mathcal{P}_i} d_{\cos}(q, s)}_{\text{MaxSim (forward)}} + \underbrace{\frac{1}{|\mathcal{P}_i|} \sum_{s \in \mathcal{P}_i} d_{\cos}(s, q)}_{\text{MeanSim (reverse)}}$$

The first term captures the *best local match* (the most relevant sentence), while the second penalizes documents with many unmatched sentences. In v1, these components are implicitly weighted equally ($\alpha = 0.5$). We investigate whether decoupling this balance can improve performance.

Mixed initial field. We replace the Chamfer-based initial concentration with an explicit mixture:

$$C_0(i) = \alpha \cdot \text{MaxSim}(q, \mathcal{P}_i) + (1-\alpha) \cdot \text{MeanSim}(q, \mathcal{P}_i)$$

where $\text{MaxSim}(q, \mathcal{P}_i) = \max_{s \in \mathcal{P}_i} \cos(q, s)$ and $\text{MeanSim}(q, \mathcal{P}_i) = \frac{1}{|\mathcal{P}_i|} \sum_{s \in \mathcal{P}_i} \cos(q, s)$. The graph topology remains Chamfer-based (preserving the high-fidelity Riemannian manifold), and the PDE solver uses pure diffusion (V4 mode, no reaction term).

Results across two BEIR datasets:

α	NFCorpus	Δ	SciFact	Δ
0.5 (≈ Chamfer)	0.2539	$+1.2\%$	0.4431	-1.5%
0.6	0.2567	$+2.3\%$	0.4434	-1.5%
0.7	**0.2588**	**$+3.2\%$**	0.4413	-1.9%
0.8	0.2578	$+2.8\%$	0.4411	-2.0%
0.9	0.2546	$+1.5\%$	0.4388	-2.5%
1.0 (pure MaxSim)	0.2535	$+1.1\%$	0.4319	-4.0%

Baselines: NFCorpus Shape-CFD v4 = 0.2508; SciFact Shape-CFD v4 = 0.4500.

The results reveal a striking *dataset-dependent divergence*:

- **NFCorpus** (heterogeneous, multi-sentence biomedical documents): The α response follows an inverted-U curve with a peak at $\alpha = 0.7$ ($+3.2\%$). MaxSim-dominated initialization penetrates the signal dilution caused by long, noisy documents, while the 30% MeanSim component provides a "macroscopic gravity" that prevents false-positive promotion.

- **SciFact** (homogeneous, short scientific abstracts): The response is *monotonically decreasing*—the higher α is, the worse the performance, with pure MaxSim ($\alpha = 1.0$) degrading by -4.0%. In 3–5 sentence documents where every sentence is signal, MaxSim $\approx$ MeanSim, so decoupling adds no information but introduces extreme-value *sampling noise* from picking a single sentence.

This asymmetry is analyzed theoretically in Section 5.4.

4.4 Ablation Study

Impact of the Shape Extension

We isolate the contribution of the shape (point cloud) representation by comparing three configurations on the BEIR datasets:

Configuration	SciFact	NFCorpus	FiQA	Description
Cosine (no PDE)	0.4701	0.2209	0.1683	No graph dynamics
AD-Rank v2 (point PDE)	0.4753	0.2398	0.1683	Graph dynamics, point embeddings
Shape-CFD (shape PDE)	**0.4820**	**0.2684**	0.1683	Graph dynamics, point clouds

The shape extension provides the marginal gain over AD-Rank v2: +1.4% on SciFact, +11.9% on NFCorpus. The FiQA null result confirms the hypothesis: the shape extension is *selectively beneficial*, activating only when documents contain multi-sentence structure. This dataset-adaptive behavior is a strength, not a limitation—it ensures no degradation on short-document corpora.

Péclet Number Sensitivity

We vary the advection strength $u_{\text{strength}} \in \{0, 0.05, 0.1, 0.2, 0.5, 1.0\}$ while fixing $D = 0.15$ on the LawVein dataset:

u_{strength}	Pe	Convergence	Blind Score	Ranking
0.00	0.000	✓	3.1	Pure diffusion
0.05	0.033	✓	3.2	Under-advected
0.10	**0.065**	**✓**	**3.4**	**Optimal**
0.20	0.130	✓	3.1	Over-advected
0.50	0.330	✗	2.8	Non-convergent
1.00	0.650	✗	2.3	Divergent

The optimal operating point is at $\text{Pe} \approx 0.065$, in the weakly advective regime. This result is robust across queries: the $u_{\text{strength}} = 0.1$ configuration achieves 100% convergence while consistently outperforming both pure diffusion ($u = 0$) and stronger advection variants.

4.5 Latency Analysis

End-to-end latency measurements on a single-core AMD EPYC (Zen 4, 4.2 GHz):

Component	AD-Rank v2	Shape-CFD
HNSW retrieval (30 candidates)	1.2 ms	1.2 ms
Sentence embedding (API)	—	1,815 ms (uncached)
JL projection + caching	—	0.3 ms (cached)
Graph construction	0.4 ms	2.1 ms
CFD solving (30-50 iterations)	1.0 ms	1.5 ms
Total (cached)	**2.6 ms**	**5.1 ms**
Total (with reranking overhead only)	**1.4 ms**	**3.9 ms**

Key observations: (1) The raw PDE solving time is negligible (≤ 1.5 ms), confirming that the algorithm's computational core is extremely efficient for the problem scale ($N = 30$). (2) The primary bottleneck for Shape-CFD is the **one-time** sentence embedding cost (1,815 ms per document via API); once cached in SQLite, subsequent queries incur only the 0.3 ms projection lookup. (3) Both methods are approximately $100\times$–$200\times$ faster than cross-encoder rerankers (500+ ms for $N = 30$).

Optimization trajectory. Over the course of development, we achieved a cumulative $300\times$ speedup through five stages of optimization:

Stage	Latency	Optimization	Factor
v0 (baseline)	450 ms	No optimization	1×
v1	120 ms	Pre-compute similarity matrix	3.75×
v2	14.8 ms	KNN sparse graph ($k=3$)	8.1×
v3	3.3 ms	Adaptive early stopping	4.5×
v4 (Shape-CFD)	22 ms	+Shape with JL ($m=128$) + cache	—

4.6 Case Study: Cross-Domain Legal Query

We present a representative query to illustrate Shape-CFD's qualitative advantage:

Query: "离婚时一方隐藏财产怎么处理" (How to handle hidden property during divorce)

This query spans two legal domains: (1) marriage law (Article 1092 of Civil Code on concealment of marital property) and (2) civil procedure law (evidence preservation provisions). A pure cosine ranker retrieves documents from both domains but fails to recognize their complementary relationship.

Cosine ranking: The top-3 results correctly include Article 1092 but rank it at position 3, behind two marginally relevant provisions from the general section of marriage law.

Shape-CFD ranking: By constructing a Chamfer-distance graph, Shape-CFD detects that Article 1092's sentence-level embedding cloud has high affinity with both the query and a procedural provision about evidence. The convection-diffusion dynamics propagate the relevance signal from the highly-relevant Article 1092 to semantically adjacent provisions, while the advection term ensures directionality: information flows preferentially from more query-aligned to less query-aligned nodes. Result: Article 1092 is promoted to position 1, with the complementary procedural provision correctly at position 2.

Why pure cosine fails: The centroid embedding of Article 1092 suffers from "semantic averaging"—the marriage-law and property-law sentence embeddings, when averaged, produce a centroid that is approximately equidistant from the query as several less relevant provisions. The Chamfer distance, by matching at the sentence level, correctly identifies the strong local alignment.

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5 Analysis: Physical Interpretation of the Optimal Regime

5.1 Why Weak Advection Outperforms Strong Advection

The empirical finding that $\text{Pe} \approx 0.065$ is optimal—a regime where diffusion dominates advection by a factor of $\sim 15\!\times$—may appear counterintuitive. If the advection term encodes query direction, why does increasing it degrade performance?

We offer a physical interpretation grounded in transient-state dynamics. The convection-diffusion equation admits two characteristic timescales:

- **Diffusive timescale:** $\tau_D \sim L^2 / D$, where L is the graph diameter
- **Advective timescale:** $\tau_U \sim L / U$, where $U = \overline{|u_{ij}|}$

The Péclet number is their ratio: $\text{Pe} = \tau_D / \tau_U = UL/D$. When $\text{Pe} \ll 1$, diffusion equilibrates much faster than advection can transport concentration directionally. Our early-stopping criterion ($\epsilon = 10^{-3}$, typically 30–50 iterations) truncates the dynamics well before the advective transport has fully developed. In this transient regime:

1. **Diffusion performs global smoothing:** The fast diffusive mode rapidly propagates the initial cosine-similarity signal across local neighborhoods, correcting isolated misrankings.
2. **Advection provides a directional perturbation:** The slow advective mode contributes a subtle, systematic bias toward query-aligned nodes, improving ranking without overwhelming the stable diffusive backbone.

When Pe is increased beyond ~ 0.1, the advective timescale becomes comparable to the diffusive timescale, and the early-stopped transient captures unstable advective oscillations before they are damped by diffusion. This manifests as convergence failure (the iteration diverges or oscillates) and degraded ranking quality.

Analogy to numerical methods. This phenomenon parallels the classical observation in CFD that explicit upwind schemes are unconditionally stable for diffusion-dominated flows ($\text{Pe} \leq 1$ locally) but require increasingly aggressive CFL restrictions or implicit methods for advection-dominated regimes. Our explicit scheme naturally operates best in the diffusion-dominated regime.

5.2 The Advection Signal in High-Dimensional Embedding Spaces

The curse of dimensionality provides a complementary explanation for weak advection. For vectors uniformly distributed on the d-dimensional unit sphere, inner products concentrate around zero: $$\langle x, y \rangle \sim \mathcal{N}\left(0, \frac{1}{d}\right)$$

With $d = 4096$, the standard deviation is $\sigma \approx 0.016$, meaning typical advection coefficients $u_{ij} \in [-0.048, +0.048]$ (within 3σ). These values are inherently small compared to the diffusion coefficient $D = 0.15$, making the weak-advection regime the *natural* operating point of the algorithm in high-dimensional spaces, rather than an arbitrary choice.

This offers a theoretical justification for the algorithm's robustness: the high dimensionality of modern embedding spaces naturally constrains the system to the stable, diffusion-dominated regime, making the algorithm self-regularizing without requiring careful tuning of the advection strength.

5.3 Limitations

We acknowledge several limitations of the current work, including three fundamental theoretical issues that we identify as priorities for future resolution:

1. **Metric space inconsistency in advection (theoretical).** As discussed in Section 3.6, Shape-CFD's graph topology is defined by the Chamfer distance (a set-theoretic metric on point clouds), while the advection field is computed from centroid vector differences (a Euclidean metric on single points). This places the diffusion and advection operators in incompatible metric spaces—the graph "knows" about fine-grained sentence matches, but the advection "sees" only the collapsed centroid. This hybrid formulation, while empirically effective, lacks rigorous differential-geometric justification and may fail on documents with highly multimodal sentence distributions.

2. **Noise dilution in symmetric Chamfer distance (practical).** The symmetric Chamfer formulation systematically penalizes long, heterogeneous documents. When a document contains K sentences but only one is relevant, the $(K-1)$ irrelevant sentences contribute large reverse-direction distances that dilute the signal. This contradicts the established asymmetric matching principle validated by ColBERT's MaxSim operator [22], and may explain Shape-CFD's limited gains on corpora with predominantly short documents (e.g., FiQA).

3. **JL projection introduces noise, not signal (theoretical).** The $O(1/m)$ variance expansion from Johnson-Lindenstrauss projection is stochastic noise injected by the random matrix, not a meaningful physical signal. The algorithm's reliance on diffusion dominance ($\text{Pe} \approx 0.065$) effectively marginalizes the advection term—including its noise components—but a PDE framework whose directional transport

mechanism is driven by projection artifacts is theoretically fragile. Scalar potential convection (Section 6) would eliminate this dependence entirely.

4. **Scale.** Our experiments use $N = 30$ candidates. Scaling to $N = 100$+ would require investigating graph connectivity and the $O(N^2)$ similarity matrix cost.

5. **Ground-truth evaluation.** The LawVein results rely on LLM blind evaluation rather than human-annotated relevance judgments.

6. **Over-smoothing without reaction terms.** The linear diffusion dynamics guarantee convergence to a flat equilibrium as $t \to \infty$. Our early-stopping strategy mitigates this practically but does not resolve it theoretically. **v3 update:** Section 4.3.3 demonstrates that the Allen-Cahn reaction term, while theoretically sound, is *numerically dormant* in the short-query regime ($\Delta C_0 \approx 0.1$) and cannot substitute for early stopping in this operating regime.

7. **Domain-dependent initial field (v3).** Section 4.3.4 demonstrates that the mixed initial field $C_0 = \alpha \cdot \text{MaxSim} + (1-\alpha) \cdot \text{MeanSim}$ yields a $(+3.2\%)$ improvement on NFCorpus at $\alpha = 0.7$ but degrades on SciFact at all $\alpha > 0.5$. This makes the mixed initial field a *domain-specific enhancement* rather than a universal improvement, limiting its applicability in zero-shot settings where corpus characteristics are unknown.

4.3.5 Subspace Chamfer Distance (v3 supplement)

A key question arising from the JL projection analysis (Section 5.3, Limitation 3) is whether alternative dimensionality strategies can improve Chamfer distance discrimination without the noise injection inherent to random projection. We investigate a *structured decomposition* approach inspired by Product Quantization [Jégou et al., 2011] and the Sparse Grid method from high-dimensional CFD [Bungartz and Griebel, 2004].

Method. We partition the $d = 4096$ dimensional embedding space into $M = 32$ contiguous subspaces of dimension $d_s = 128$. For each subspace m, we compute an independent cosine distance $d_m(a, b) = 1 - \cos(a_{[m]}, b_{[m]})$, where $a_{[m]}$ denotes the m-th 128-dimensional slice of vector a. The aggregate PQ-Chamfer distance is defined as the mean over subspaces:

$$d_{\text{PQ}}(a, b) = \frac{1}{M} \sum_{m=1}^{M} d_m(a_{[m]}, b_{[m]})$$

This decomposition preserves the original dimensional structure—unlike JL projection, which stochastically mixes all dimensions via a random Gaussian matrix. Each 128-dimensional subspace exhibits higher distance variance than the full 4096-dimensional space (the *concentration of measure* effect weakens as d decreases), thereby improving the discriminative power of Chamfer distance.

Results (NFCorpus, 323 queries):

Method	NDCG@10	Δ vs V4
Shape-CFD V4 (4096d cosine distance)	0.2508	—
Shape-CFD + JL128 (random projection to 128d)	0.2316	−7.6%

Method	NDCG@10	Δ vs V4
Shape-CFD + PQ-Chamfer (32×128 subspaces)	0.2684	+7.0%
Shape-CFD + PQ-Chamfer (64×64 subspaces)	**0.2735**	**+9.1%**

An ablation study over the subspace granularity reveals a monotonic improvement with finer partitions: 16×256 (0.2650) → 32×128 (0.2684) → 64×64 (0.2735), consistent with the $O(1/\sqrt{d})$ distance variance scaling. Variance-weighted aggregation (weighting subspaces by query energy) degrades performance by −3.7%, indicating that Qwen3-Embedding distributes semantic information uniformly across dimensions. Dimension permutation has negligible effect (−0.1%), confirming that the embedding dimensions lack local semantic structure.

The PQ-Chamfer decomposition achieves a +9.1% improvement over the full-dimensional baseline, compared to the −7.6% degradation of JL projection—a total swing of 16.7 percentage points. This confirms that the critical factor is not dimensionality reduction per se, but whether the dimensional transformation *preserves or destroys* the embedding structure.

Physical interpretation. In the CFD analogy, this corresponds to replacing a *uniform coarse mesh* (full 4096d cosine, low resolution due to concentration) with a *multi-block structured mesh* (32 independent 128d domains, each with higher effective resolution). The subspace distances act as independent "sensors" that detect fine-grained semantic differences invisible to the full-dimensional metric, analogous to Adaptive Mesh Refinement (AMR) applied to different regions of the embedding space.

5.4 Subspace Virtual Tokens: PQ Decomposition as Semantic Factorization (v4 supplement)

The PQ-Chamfer distance introduced in Section 4.3.5 decomposes the 4096-dimensional embedding space into 64 contiguous 64-dimensional subspaces and computes independent cosine distances within each, yielding a +8.8% improvement over full-dimensional Chamfer. In this v4 supplement, we discover that the **order of operations**—subspace aggregation versus sentence-level Chamfer—critically affects retrieval quality, and that reversing the standard order unlocks an additional +2.5% improvement at zero computational cost.

The Operation Order Insight

Standard PQ-Chamfer (Section 4.3.5) first aggregates across 64 subspaces to produce a single scalar distance between two vectors, then applies the Chamfer operator across sentences:

$$d_{\text{PQ-Chamfer}}(q, D) = \text{Chamfer}\left(\{q\}, D; \; d_{\text{PQ}}\right)$$

VT-Aligned reverses this order: for each subspace s independently, it first computes the sentence-level Chamfer distance (in 64 dimensions), then averages across subspaces:

$$d_{\text{VT}}(q, D) = \frac{1}{S} \sum_{s=0}^{S-1} \left[\min_{j \in D} d_s(q, s_j) + \frac{1}{|D|} \sum_{j \in D} d_s(s_j, q) \right]$$

where $d_s(a, b) = 1 - \cos(a^{(s)}, b^{(s)})$ is the cosine distance restricted to the s-th 64-dimensional subspace, and $a^{(s)}$ denotes the s-th subspace slice of vector a.

Physical interpretation. Each 64-dimensional subspace acts as an independent "virtual token"—a semantic dimension that captures one facet of the full embedding. The query's single 4096d vector is effectively decomposed into 64 independent semantic probes, each searching for its best

match across all document sentences in its own 64-dimensional subspace. This increases the effective query density from 1 point to 64 points without any model re-inference.

We also evaluate an **unaligned** variant (VT-Unaligned) that flattens all subspaces into a single pool of 64d virtual tokens and computes full Chamfer in 64d space, allowing cross-subspace matching (subspace i of the query can match subspace $j \neq i$ of a document sentence).

Stacking with Adjoint Prefetch

We combine VT-Aligned with the V7.1 adjoint state prefetch (Section 5.5), using VT-Aligned distance for graph construction and PDE initial conditions, and the adjoint flux formula for boundary node selection: $$i^* = \arg\max_{i \in \partial V} C_i \cdot \max\left(0, \sum_{j \in \mathcal{N}(i)} U_{ij} W_{ij} C_j\right)$$

Probe vectors are constructed via cosine interpolation ($\text{probe} = 0.7 \hat{q} + 0.3 \hat{d}_i$) and retrieved via cosine similarity over the full corpus.

Results (NFCorpus, 323 queries)

Method	NDCG@10	vs Cosine	vs PQ-Chamfer
Cosine baseline	0.2195	—	—
PQ-Chamfer (Section 4.3.5)	0.2729	+24.3%	baseline
VT-Unaligned	0.2753	+25.4%	+0.9%
V7.1 Adjoint Prefetch	0.2759	+25.7%	+1.1%
VT-Aligned	**0.2797**	**+27.4%**	**+2.5%**
VT-Aligned + V7.1 Adjoint (VT-V7)	**0.2844**	**+29.6%**	**+4.2%**

Key findings:

1. **VT-Aligned > VT-Unaligned** (+0.9% vs +2.5%): Cross-subspace matching introduces noise, consistent with the LID analysis (Section 4.3.5) showing Qwen3-8B embeddings distribute information uniformly across dimensions.

2. **VT-Aligned + V7.1 stack near-orthogonally**: Individual gains are +2.5% and +1.1%; combined gain is +4.2%, exhibiting slight superlinear stacking. This confirms the two innovations address independent bottlenecks (distance granularity vs. candidate pool expansion).

3. **Zero additional cost**: VT-Aligned requires no model re-inference, no additional data, and no new hyperparameters. The only change is the order of mathematical operations on existing sentence vectors.

Rust Acceleration

We implement the full VT-V7 pipeline (cosine pre-filter → VT-Aligned graph construction → PDE solving → adjoint prefetch) as a native Rust module with NAPI-RS bindings, achieving **38 ms per query** on commodity CPU hardware—a 130× speedup over the JavaScript reference implementation. The Rust module (`law-vexus`) includes 45 unit tests and uses `rayon` for parallel distance matrix computation on multi-core systems.

5.5 The Pareto Boundary of Initial Fields (v3 supplement)

The dataset-dependent behavior of the mixed initial field (Section 4.3.4) admits a precise statistical interpretation. We formalize this as the **Dual-Role Theorem of MeanSim**.

Theorem 2 (Dual-Role Theorem). *In the mixed initial field $C_0 = \alpha \cdot \text{MaxSim} + (1-\alpha) \cdot \text{MeanSim}$, the MeanSim component plays two distinct roles depending on document structure:*

(i) Low-pass filter (short, homogeneous documents). When $K_i \leq K_{\text{crit}}$ (few sentences, all topically coherent), $\text{MaxSim} \approx \text{MeanSim}$ and the decoupling adds no semantic information. However, MaxSim introduces extreme-value sampling noise (variance $\sim 1/K_i$), while MeanSim provides variance reduction via averaging. Increasing α degrades performance monotonically by amplifying noise without signal gain.

(ii) Macroscopic gravity (long, heterogeneous documents). When $K_i \gg K_{\text{crit}}$ (many sentences with sparse relevance), MeanSim suffers catastrophic signal dilution ($\text{MeanSim} \to 0$ as $K_i \to \infty$ for fixed relevant sentences). MaxSim penetrates this dilution by isolating the highest-scoring sentence. The optimal α^* balances the signal gain from MaxSim against its false-positive risk, yielding the inverted-U response observed on NFCorpus.

The transition between regimes is governed by the *effective signal-to-noise ratio* of the MaxSim operator:

$$\text{SNR}_{\text{MaxSim}} = \frac{\mathbb{E}[\max_s \cos(q, s_{\text{rel}})] - \mathbb{E}[\max_s \cos(q, s_{\text{irrel}})]}{\text{Var}[\max_s \cos(q, s)]^{1/2}}$$

When $\text{SNR}_{\text{MaxSim}} > 1$ (long documents with strong local evidence), decoupling is beneficial; when $\text{SNR}_{\text{MaxSim}} < 1$ (short documents where the extreme value is noise-dominated), the symmetric Chamfer baseline ($\alpha = 0.5$) is optimal.

Scale-adaptive symmetry breaking. Based on this analysis, we propose a document-level adaptive mixing coefficient:

$$\alpha_i = 0.5 + \Delta_{\max} \cdot \tanh\left(\max\left(0, \frac{|D_i| - L_{\min}}{L_{\text{scale}}}\right)\right)$$

where $|D_i|$ is the sentence count of document i, $L_{\min}$ is the minimum length below which $\alpha = 0.5$ (the Chamfer default), L_{scale} controls the transition steepness, and $\Delta_{\max} \in [0.2, 0.25]$ caps the maximum asymmetry. This formula satisfies two critical properties: (1) it reduces to the v1 Chamfer baseline ($\alpha \equiv 0.5$) for corpora of uniformly short documents, ensuring no degradation; (2) it smoothly transitions to the $\alpha = 0.7$–0.75 regime for long-document corpora, capturing the NFCorpus-type improvement. Experimental validation of this adaptive formula is deferred to future work.

5.5 Speculative Retrieval: From Reranker to Retriever (preliminary)

A fundamental limitation of reranking-based approaches is the **candidate ceiling**: the reranker can only reorder documents already present in the initial retrieval set. If a relevant document is ranked beyond the top-N cutoff by the initial cosine retriever, no amount of reranking can recover it.

We propose **Speculative Retrieval**, inspired by CPU branch prediction: after an initial PDE pass on the top-N candidates, we identify *high-pressure boundary nodes*—documents with high post-PDE concentration but low graph degree (i.e., at the "edge" of the semantic neighborhood). These nodes indicate that the candidate set boundary intersects a region of high relevance, and that additional relevant documents may exist just beyond the retrieval horizon.

For each high-pressure boundary node, we compute a *diffusion probe* vector by interpolating between the node's embedding and the query direction, then retrieve the nearest neighbors of this probe from the full corpus. The expanded candidate set is then subjected to a second PDE pass.

Preliminary results (NFCorpus, 323 queries):

Method	NDCG@10	Δ vs PQ64-top30
PQ-Chamfer top-30 (baseline)	0.2729	—
Naive expansion to top-40	0.2766	+1.4%
Naive expansion to top-50	0.2757	+1.0%
Speculative Retrieval	**0.2781**	**+1.9%**

The speculative approach outperforms naive pool expansion (+1.9% vs +1.4%), confirming that the PDE concentration field provides directional information beyond what cosine similarity alone captures. This represents a preliminary step toward upgrading Shape-CFD from a pure reranker to a retrieval-augmented reranker that can actively discover relevant documents missed by the initial retriever.

6 Conclusion

We have presented Shape-CFD, a physics-inspired reranking framework that models post-retrieval reranking as a conservative convection-diffusion process on a document similarity graph. By introducing directional advection alongside isotropic diffusion, and by upgrading document representations from single-point embeddings to sentence-level point clouds measured via Chamfer distance, Shape-CFD captures both inter-document structural relationships and fine-grained sub-document semantic matches.

v4 supplement findings. The VT-Aligned virtual token decomposition (Section 5.4) demonstrates that PQ subspace structure can be repurposed as a semantic factorization tool: by computing sentence-level Chamfer within each 64-dimensional subspace independently, the effective query representation density increases 64-fold without model re-inference. Combined with the adjoint state prefetch (Section 5.5), the VT-V7 configuration achieves NDCG@10 = 0.2844 (+29.6% over cosine), establishing a new all-time best. A Rust implementation with SIMD-friendly memory layout and `rayon` parallelism reduces end-to-end latency to 38 ms per query—a 130× speedup over the JavaScript reference.

v3 supplement findings. The systematic ablation of V5 reaction-diffusion extensions (Section 4.3.3) reveals that the Allen-Cahn reaction term is physically dormant in the short-query, narrow score-range regime—not due to implementation deficiency, but as a consequence of the cubic scaling of reaction forces versus linear scaling of diffusion in the $\Delta C_0 \ll 1$ limit. The mixed initial field experiment (Section 4.3.4) demonstrates that decoupling the symmetric Chamfer distance into explicit MaxSim and MeanSim components yields a domain-dependent response: a $+3.2\%$ improvement on NFCorpus (heterogeneous long documents) at $\alpha = 0.7$, but monotonic degradation on SciFact (homogeneous short documents). This Pareto boundary (Section 5.4) is governed by the trade-off between extreme-value signal gain and

sampling noise, and the original symmetric Chamfer ($\alpha = 0.5$) remains the Pareto-optimal default for domain-agnostic deployment.

Future Work

The three theoretical limitations identified in Section 5.3 point directly to a coherent next-generation design. We outline the key directions:

1. **Scalar potential convection (resolves Limitations 1 and 3).** The metric space inconsistency and JL noise dependence share a common root cause: defining advection via vector inner products in high-dimensional space. We propose replacing the centroid-based advection with a *scalar potential gradient*, following the Darcy flow analogy: $$u_{ij}^{\text{potential}} = \kappa \cdot (\Phi_j - \Phi_i)$$ where $\Phi_i = \max_{s \in \mathcal{P}_i} \cos(q, s)$ is the MaxSim score of document i against the query. This formulation eliminates the curse of dimensionality entirely—scalar subtraction has no dependence on the ambient dimension d—and unifies the metric space: both the graph topology and the advection field operate on the same score manifold. The resulting advection is guaranteed to be non-degenerate whenever two documents have different MaxSim scores, regardless of the embedding dimension.

2. **Allen-Cahn reaction-diffusion for long-document retrieval (resolves Limitation 6, v3 update).** Section 4.3.3 empirically confirms that the reaction term is dormant in the short-query regime ($\Delta C_0 \approx 0.1$). However, in long-document retrieval scenarios (e.g., legal case matching, multi-hop QA), where $\Delta C_0 \gg 0.1$ due to sparse evidence distributed across lengthy documents, the Allen-Cahn dynamics are expected to activate meaningfully. The domain-adapted formulation $\gamma_{\text{eff}} = \gamma_{\text{base}} / (\Delta C)^2$ (Section 4.3.3) provides the correct scaling law for such scenarios. We propose augmenting the convection-diffusion equation with the domain-adapted reaction term: $$R(C_i) = \gamma_{\text{eff}} \, (C_i - C_{\min})(C_{\max} - C_i)(C_i - \theta_{\text{Otsu}})$$ where θ_{Otsu} is dynamically determined via Otsu's method. Target benchmarks include COLIEE (legal case matching) and TREC-Robust04 (narrative long queries).

3. **Asymmetric shape matching (resolves Limitation 2).** Replacing the symmetric Chamfer distance with a directed, query-weighted variant: $$d_{\text{asym}}(\mathcal{P}_q, \mathcal{P}_i) = \frac{1}{|\mathcal{P}_q|} \sum_{a \in \mathcal{P}_q} \min_{b \in \mathcal{P}_i} d_{\cos}(a, b)$$ This measures only whether the query's semantic facets are *covered* by the document, without penalizing unmatched document sentences. Combined with LSH (locality-sensitive hashing) sign quantization and hardware popcount instructions, this could achieve sub-millisecond shape distance computation.

4. **Spectral acceleration.** For larger candidate sets ($N \geq 100$), evolving the concentration field in the spectral domain of the graph Laplacian could reduce the iteration count from $O(N)$ to $O(\log N)$.

5. **Scale-adaptive initial field (resolves Limitation 7, v3).** The Pareto boundary analysis (Section 5.4) motivates a document-level adaptive mixing coefficient $\alpha_i = 0.5 + \Delta_{\max} \cdot \tanh(\max(0, (|D_i| - L_{\min}) / L_{\text{scale}}))$ that automatically transitions from the symmetric Chamfer baseline ($\alpha = 0.5$) for short documents to the asymmetric MaxSim-dominant regime ($\alpha \approx 0.7$) for long documents. Combined with per-query calibration via the MaxSim SNR criterion, this would yield a fully adaptive, domain-agnostic initial field.

6. **Chunk-level graph construction for long-document retrieval.** For corpora with documents exceeding hundreds of sentences (e.g., legal case files, clinical trial reports), document-level graph construction loses the fine-grained Riemannian manifold structure that underpins Shape-CFD's success. We propose segmenting long documents into semantic chunks and constructing the KNN graph at the chunk level, creating a much larger but topologically richer graph where the Allen-Cahn reaction term (item 2) is expected to become physically active due to the expanded score range ($\Delta C_0 \gg 0.1$).

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